

Number of Provinces (Union-Find Code)

```
int findCircleNum(vector<vector<int>>& isConnected) {
    //n x n matrix
    int n = isConnected.size();
    vector<int> par, rank;
    for(int i=0; i<n; i++) {
        par.push_back(i);
        rank.push_back(0);
    }

    for(int i=0; i<n; i++) {
        for(int j=0; j<n; j++) {
            if(isConnected[i][j] == 1) {
                unionByRank(i, j, par, rank);
            }
        }
    }

    set<int> s = {};
    for(int p: par) {
        s.insert(find(p, par, rank)); //not to add par, but set leader
    }

    return s.size();
}

int find(int x, vector<int> &par, vector<int> &rank) {
    if(x == par[x]) {
        return x;
    }

    return par[x] = find(par[x], par, rank);
}

void unionByRank(int a, int b, vector<int> &par, vector<int> &rank) {
    int parA = find(a, par, rank);
    int parB = find(b, par, rank);

    if(parA == parB) return;
```

```
if(rank[parA] == rank[parB]) {  
    par[parB] = parA;  
    rank[parA]++;  
} else if(rank[parA] > rank[parB]) {  
    par[parB] = parA;  
} else {  
    par[parA] = parB;  
}  
}
```